MIL-T-83956A (USAF)
17 January 1977
SUPERSEDING
MIL-T-83956 (USAF)
29 JULY 1970

MILITARY SPECIFICATION

TOWER, ASR-4, 10 FOOT AND 17 FOOT SECTIONS

This specification is approved for use by Air Force, Department of the Defense, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of tower designated, Tower, ASR-4 and is hereinafter designated as the tower.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

WW-P-404	Pipe, Steel, Seamless and Welded, Black and Zinc Coated Galvanized
* FF-B-575	Bolts, Hexagon and Square
RR-G-661	Grating, Metal Floor (Except for Naval vessels)
FF-N-836	Nuts, Square, Hexagon, Cap, Slotted, Castle, Knurled, Welding and Single Ball Seat

MILITARY

MIL-P-116	Preservation, Methods of
MIL-C-6136	Conduit, Electrical, Flexible, Shielded Aluminum Alloy, for Aircraft Installations

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Warner Robins ALC/ MMIRDA, Robins AFB GA 31098 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-R-11461	Rod, Ground (with attachments)
MIL-M-17194	Metal, Expanded, Steel

STANDARDS

Military

MIL-STD-129	Marking for Shipping and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specifications and Standards, Order of Procedure for Selection of
MIL-STD-163	Steel Mill Products, Preparation for Shipment and Storage
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
* MIL-STD-882	Systems Safety Program for Systems, Associated Sub Systems and Equipment; Requirements for
* MIL-STD-889	Dissimilar Metals

DRAWINGS

Federal Aviation Agency, Bureau of Facilities, Plant Engineering Branch

For a 17 Foot Top Section	For a 10 Foot Bottom Section
D5453-E15	
D5453-E16	D5453-E13
D5453-E1	D5453-E14
D5453-E8	D5454-1
D5453-E9	D5454-2
D5453-E10	D5454-3
D5453-E11	D5454-4
D5453-E12	D5454-5
D5454-6	D5419-1
D5454-7	D5419-5
D5454-8	D5419-6
D5454-9	D5419-9
D5454-10	
D5454-11	
D5454-12	
D5454-12A	
D5454-13	
D5454-14	
D5454-15	
D5454-16	
D5454-17	

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D5454-18

D5419-1
D5419-2
D5419-3
D5419-4
D5419-5
D5419-6 (For Reference Only) 10' Ext. Section
D5419-7 (For Reference Only) Field Alterations
D5419-8 (For Reference Only) Elec. Installation
D5419-9 (For Reference Only) Foundation Plan and Details

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

A-36	Specification for Structural Steel
A-123	Zinc (Hot Dip) Coating on Products Fabricated from Rolled, Pressed and Forged Steel Shapes, Plates, Bars, and Strips
A-153	Zinc Coating (Hot Dip) on Iron and Steel Hardware
A-325	Specification for High Strength Carbon Steel Bolts, Including Suitable Nuts, and Plain Hardened Washers
A-449	Specifications for Quenched and Tempered Steel Bolts and Studs
A-490	Specification for Quenched and Tempered Alloy Steel Bolts for Structural Steel Joints

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

* AMERICAN WELDING SOCIETY

Welding Code D1.1

(Applications for copies should be addressed to the American Welding Society, Inc. (AWS) 2501 N W 7th Street, Miami FL 33125.)

UNIFORM CLASSIFICATION COMMITTEE (UCC)

Uniform Freight Classification Rules

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(Applications for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 W Jackson Blvd, Chicago IL 60606.)

* DEPARTMENT OF LABOR OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION

Occupational Safety and Health Act (OSHA) Standards with Supplemental Revisions.

(Applications for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington D C 20402.)

NATIONAL MOTOR FREIGHT CLASSIFICATION RULES

(Applications for copies should be addressed to the American Trucking Association Inc (ATA), Traffic Department, 1616 P Street, N.W. Washington D.C. 20036.)

3. REQUIREMENTS

3.1 Preproduction testing. This specification makes provision for preproduction testing.

3.2 Description, general. The tower shall be composed of one 17 foot section, and one to six 10 foot sections, giving the tower a height of from 17 feet to 77 feet.

3.3 Materials, parts, and processes. Materials, parts, and processes shall be as specified herein. Materials and parts not covered by applicable specifications shall be of the best commercial quality, the lightest practical weight, entirely suited for the purpose, and readily available.

3.3.1 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

* 3.4 Dissimilar metals. Dissimilar metals as defined by MIL-STD-889 shall not be used in intimate contact, without suitable protection against electrolytic corrosion.

* 3.5 Materials.

3.5.1 Structural steel. Steel for structural and framing members shall be in accordance with ASTM Standard A-36. All structural steel shall be finished as specified herein (see 3.10).

* 3.5.2 Fasteners. Bolts for joining main structural members shall be high strength bolts conforming to ASTM Standards A-325, A-449 or A-490 and shall be furnished with suitable nuts, palnuts or lockwashers and plain hardened washers. All other bolts shall be in accordance with FF-B-575

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and nuts in accordance with FF-N-836. All bolts, nuts, washers and other hardware shall be galvanized in accordance with ASTM designation A-153.

3.5.3 Conduit. Conduit as shown on referenced drawings shall be furnished and shall be in accordance with MIL-C-6136.

3.5.4 Structural pipe and tubing. Pipe and tubing as shown on referenced drawings shall be in accordance with WW-P-404.

3.5.5 Grating.

3.5.5.1 Bar type. Bar type grating shall be furnished as shown on referenced drawings and shall be in accordance with RR-G-661.

3.5.5.2 Expanded metal type. Expanded metal type grating shall be furnished as shown on referenced drawings and shall be in accordance with MIL-M-17194.

3.5.6 Extra material. The supplier shall furnish not less than an additional 5 percent of bolts, nuts, washers, and other fastenings. This additional material shall be of the same type and character supplied for the assembly of the towers in the field.

3.6 Ladder and safety rails.

3.6.1 Ladder. A steel ladder shall be furnished with each tower section by the supplier. The ladder shall be in accordance with OSHA Standard, paragraph 1910.27 Fixed Ladders.

3.6.2 Safety rails. Safety rails to be mounted on the ladder shall be furnished with the tower. Safety rails shall have provisions to receive a safety belt clevis which immediately locks upon taking the suspended weight of a man. The safety rail shall meet the requirements of OSHA Standard paragraph 1910.27(d)(5) Ladder safety devices. All necessary hardware shall be provided with the safety rail so that it may be mounted to the rungs of the ladder in accordance with the manufacturer's instructions. At least two leather belts with the required sleeves and clevises for two complete safety belts shall be furnished with each tower. The Safety System, including safety rail and safety belts, shall be designed to support a 200 pound man with a factor of safety of 3.

* 3.7 Ground rod. Two 5/8" X 8' 0" ground rods with connecting conductors in accordance with MIL-R-11461 shall be furnished with each tower.

* 3.8 Welding. All welding shall be done in accordance with the American Welding Society Structural Welding Code D1.1, and all welders shall be certified in accordance with this code. There shall be no welding, riveting or other special process required for field erection.

3.9 Service conditions.

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3.9.1 Service conditions (climatic). The tower has been designed for, and shall be constructed to withstand any probable combination of the following climatic conditions without any mechanical or electrical damage, or any degradation in the performance specified herein:

a.	Ambient temperature:	-54°C to +70°C.
b.	Relative humidity:	Up to 100 percent including condensation due to temperature changes.
c.	Barometric pressure:	From 30 to 5.54 inches of Mercury (from sea level to 40,000 feet above).
d.	Wind and ice:	85 knots per hour including gusts, with 1½ inches of radial ice operating and 130 knots per hour including gusts and 1½ inches of radial ice, non-operating.
e.	Salt atmosphere:	As encountered in coastal regions and during ocean transportation.
f.	Sand and dust:	As encountered in arid regions (particles as small as 0.0017 inch).
g.	Insects and fungi:	As encountered in the tropics.

3.9.2 Service conditions (mechanical). The equipment shall be constructed so that no fixed part will become loose, no movable part or permanently set adjustment will be shifted in setting or position and no degradation be caused in the performance specified herein.

3.9.3 Service life. The equipment shall be constructed for a minimum life expectancy of 10 years after a possible maximum storage period of 2 years.

3.10 Finish for structural steel.

3.10.1 Galvanized finish. All structural and framing members shall be galvanized by the hot dip method in accordance with ASTM designation A-123, after as much fabrication as possible is complete.

* 3.11 Interchangeability. Like units, assemblies, sub-assemblies, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment. Reliance shall not be placed on any unspecified dimension, rating, etc.

* 3.12 Identification markings.

3.12.1 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.12.2 Identification of members. All structural steel shall be depth stamped so as to permit ready identification by reference to erection and assembly drawings provided with the tower. The erection marks shall be stamped into the steel before galvanizing and shall be legible after galvanizing to the naked eye at a distance of 6 feet.

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* 3.13 Workmanship. All members and parts of the tower shall be in accordance with this specification. All shearing, punching, back and end cuts shall be accurately and carefully performed. Holes shall be true and square with the plane of the material, and all exposed raw edges, including edges exposed in the dismantled stage, shall be ground, machined, or otherwise smoothed to prevent accidental cutting or scratching of handlers and users of the tower.

3.14 Tolerances. The supplier is responsible for providing tolerances in accordance with the drawings and industrial standards to insure proper fit, assembly, etc.

3.15 Safety.

3.15.1 Category III and IV hazards. All equipment furnished under this specification shall be analyzed for Category III and IV hazards as defined by MIL-STD-882. Any hazard so identified shall be brought to the attention of the procuring activity with recommended action for satisfying safety requirements. Action order of procedence shall be as specified in MIL-STD-882 Para 5.6.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. Tests shall be classified as preproduction (see 4.4) and acceptance (see 4.5).

4.3 Test conditions. Unless otherwise specified, all tests required by this specification shall be made at prevailing ambient temperature and humidity conditions.

4.4 Preproduction test. One tower, or towers, representative of production equipment, and various components as required, shall be subjected to the tests specified herein. The samples shall be identified with the manufacturer's part number and such other information as is required by this specification.

4.4.1 Pre-erection inspection tests. The tower shall be given a thorough mechanical and visual inspection and test to determine that the quality of all materials and workmanship are in compliance with the requirements of this specification. Particular attention shall be given to the following:

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a. Choice and quality of materials.
b. Nameplates, identification markings and labels.
c. Finishes.
d. Welded joints.
e. Check of mounting means.
f. Check of corrosion prevention.
g. Check of safety features.
h. Loose fastening and securing devices or parts.
i. Grounding connections.
j. Over-all dimensions check.
k. Other visual defects.
l. Workmanship (see 3.13).

4.4.2 Assembly. The contractor shall assemble one complete tower of each class specified to insure a proper fit of all members.

4.4.2.1 Post-erection inspection tests. An erected tower shall be subjected to a careful mechanical inspection to insure that the requirements of this specification have been met. Such inspection shall demonstrate the accuracy of fit and the correctness of dimensional data. All sections shall fit easily without having to be forced and without bending or straining of any part. Particular attention shall be given to the following points:

a. The fit of members and parts in their respective positions.
b. Accessibility of parts for servicing and inspection.
c. Check of mounting means.
d. Check of safety features.
e. Nameplate.
f. Correctness of dimensional data.
g. Accuracy of shearing, punching, back and end cuts.

4.5 Acceptance tests. The acceptance tests shall consist of those tests listed in 4.4.1.

4.5.1 Preparation for delivery inspection. The preservation, packaging, packing and marking of the tower shall be inspected to verify conformance to the requirements of section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 The preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Methods of preservation and packaging for tower components and accessories shall be determined and accomplished in accordance with MIL-P-116 and MIL-STD-794. Interior containers shall be determined and utilized in accordance with MIL-STD-794. The contents of each package

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shall contain items and quantities, which will facilitate efficient and expeditious placement at the erection site for timely unpacking and erection in established steps of succession called for in the erection manual.

5.1.1.2 Fasteners. Bolts, nuts, washers, clips, screws, and similar items of like description shall be preserved in accordance with MIL-P-116, Method IC-2, except where small quantities (2 pounds or less) are involved, then Method IC-3 may be used.

5.1.1.3 Drawings and technical publications. The drawings and technical publications required for each tower shall be preserved in accordance with MIL-P-116, Method IC-1 or IC-3.

5.1.2 Level C. The complete tower shall be preserved and packaged in accordance with the supplier's standard practice.

5.2 Packing. The packing shall be A or C, as specified (see 6.2).

5.2.1 Level A. Packing and exterior shipping containers shall conform to the requirements of MIL-STD-794, except that large bar (framing) sheet, plate and structural shape components shall be packed in accordance with MIL-STD-163. The final arrangement of shipping containers shall be in accordance with MIL-STD-794 and lifts conforming to MIL-STD-163 shall be entirely suitable for identification, documentation, and accountability of the packages belonging to one (1) tower, when multiple quantities of like towers are being processed or handled simultaneously through shipping and receiving ports.

5.2.2 Level C. The complete tower shall be packed in a manner to insure arrival at destination in satisfactory condition and which will be acceptable to carriers at lowest rates. Packing shall comply with UCC Uniform Freight Classification Rules or National Motor Freight Classification Rules, according to the mode of transportation.

5.3 Marking. Interior packages, bundles, and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The tower covered by this specification is a sectionalized tower and is intended for use as an airport surveillence radar tower.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Level of preservation and packaging (see 5.2).
c. Level of packing (see 5.3).

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* 6.3 Drawing changes. In the event it becomes necessary to change the drawings in order to achieve the performance requirements of this specification, the contractor will immediately notify the contracting officer with recommendations for drawing changes to obtain the required performance. To the extent approved, such changes will be processed in accordance with the provisions of the changes clause of the contract.

* 6.4 Asterisks. The margins of this specification are marked with an asterisk * to indicate where changes (additions, modifications, corrections, deletions, etc.) from the previous issues were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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Preparing Activity:
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DOCUMENT IDENTIFIER AND TITLE

MIL-T-83956A (USAF) Tower, ASR-4, 10 Foot and 17 Foot Sections

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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